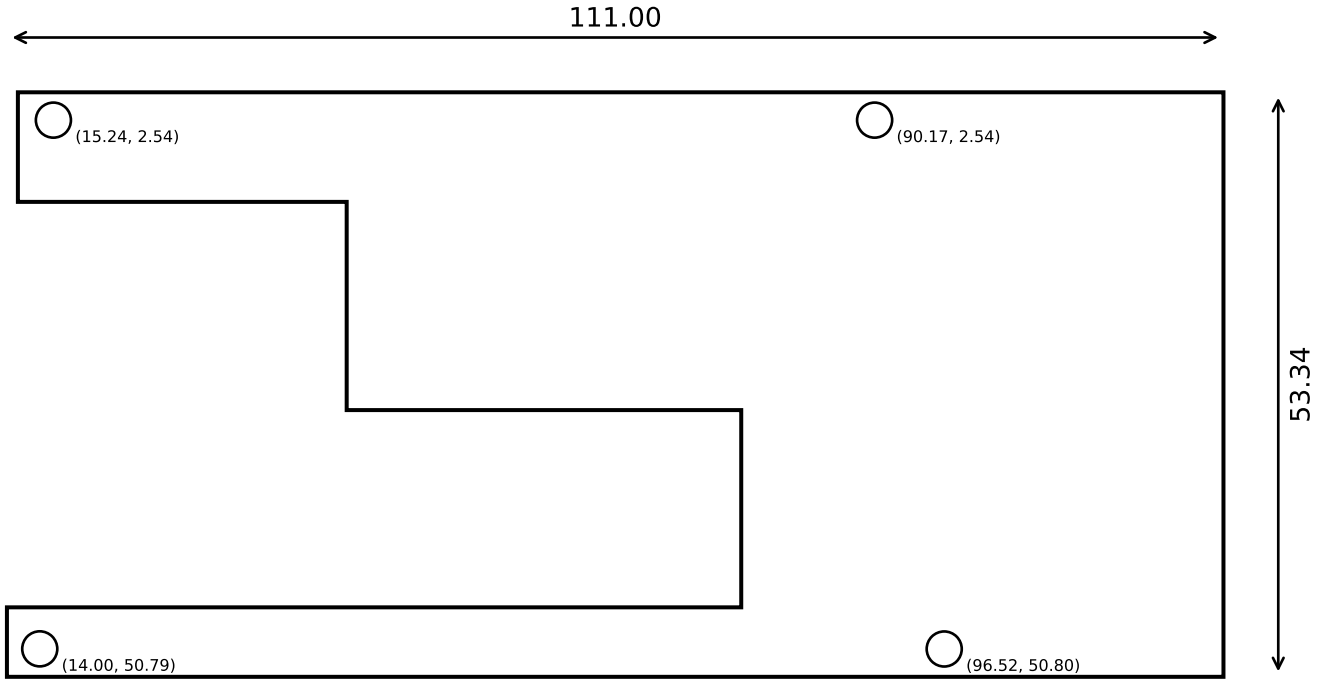
giga-adapter — fabrication drawing



Finished thickness: 1.6 mm

4 layers, FR-4

1 oz copper, all layers

ENIG

Green mask / white silk

IPC Class 2

Electrical test: 100%

Layer order: F.Cu / In1.Cu / In2.Cu / B.Cu. Routed Edge.Cuts and drill files are authoritative. Nominal stackup is embedded in the KiCad board; equivalent FR-4 laminate construction is acceptable at the specified finished thickness.

Minimum track and copper clearance 0.15 mm; through via drill 0.30 mm, diameter 0.65 mm, annular ring 0.15 mm. Tented vias both sides, no via-in-pad. Preserve fine-pitch solder-mask webs (minimum 0.13 mm).

PCBWay to add assembly rails/panelization without changing individual outlines or mounting coordinates. Do not mirror bottom fabrication files. PTH and NPTH files are separate.

Origin in Gerbers, drills and placement: X=0, Y=lower board edge; positive Y upward. Drawing coordinates above use KiCad top-left Y-down for mechanical reference.

Stencil: stepped 0.15 mm at FH28 and 0.10–0.12 mm at 0402/TSSOP parts. Hirose paste apertures are already reduced in the supplied paste layers. No substitutions for connector suffixes or header tail lengths.

giga-adapter — assembly instructions

Use assembly-F.pdf and assembly-B.pdf for reference designators, pin numbers and component orientation. Bottom drawing is mirrored to show the actual bottom view. Native KiCad placement rotations are retained in placement.csv.

All components in bom-pcbway-smt.csv are FIT. No DNP SMT components. Parts marked HAND in bom-all.csv are supplied/installed separately and must not appear in the SMT placement file.

Inspect every 0.5 mm connector contact for bridges, insufficient solder and coplanarity. Check soldermask webs and that no solder or un-tented via protrudes into the Hirose underside/mating keepout.

J1 FH28H is TOP mounted, cable mouth facing the right board edge. Pin 1 is at the lower end when the board is viewed from the top. Check the silkscreen 1/80 markings.

J10 Phoenix 1751248: HAND solder, wire entry faces left into the central opening. Pin 1 = 24V INPUT, pin 2 = GND. U8 TSR 1-24120 is also HAND soldered; pin 1 = protected 24V, pin 2 = GND, pin 3 = 12V.

U7 thermal pad and RTN belong to their own copper island; never solder-bridge them to GND. D1 is bidirectional; D2/D3 cathodes follow their band and pad 1.

J2–J9 stacking headers are HAND soldered, sockets above the PCB and tails below. J2–J8: Samtec SSQ -03, 10 mm tails. J9: SSQ-112-04-G-D, 14.83 mm tails. Verify exact suffixes in the full BOM.

GIGA top to carrier underside: 13.0 mm. Lower two mounting positions use M2 insulating support hardware; upper holes are alignment clearances without screw heads. Support the cable end independently.

Do not bridge isolated pass-through contacts to adapter ground or power. Motor/EtherCAT contacts remain accessible through the stacking sockets and are electrically unconnected on this PCB.

First-board power, timing, touch and supply-sequencing tests are listed in README.md. They have not been performed.